Academic Year: 2026-27

Subject: MDM- Wireless & Mobile Communication

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Experiment No.:	03
Title:	Demonstration of Frequency Reuse Concept in Cellular Systems (Simulation/Visualization)
Date of Performance:	
Date of Submission:	
Marks:	
Sign of Faculty:	

Aim: Demonstration of Frequency Reuse Concept in Cellular Systems (Simulation/Visualization)

Objective: The objectives of this experiment are:

1. To understand the concept of frequency reuse in cellular communication systems.
2. To visualize the arrangement of cells using a hexagonal cellular model.
3. To understand how the same frequency channels can be reused in geographically separated cells.
4. To observe the effect of co-channel interference when the same frequency is assigned to adjacent cells.
5. To study the relationship between frequency reuse distance, cluster size, and interference.

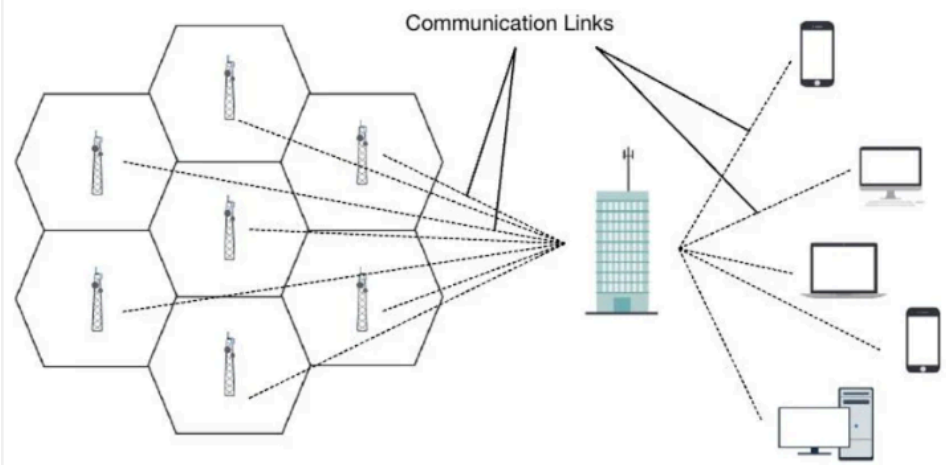
Theory:

Cellular Communication System

A cellular communication system divides a large geographical area into a number of smaller regions called **cells**.

Each cell is generally represented by a **hexagon** for theoretical and simulation purposes. A **Base Station (BS)** is located approximately at the center of each cell and provides wireless communication services to users within that cell.

Instead of assigning a unique frequency to every cell, a cellular system divides the available frequency spectrum into several groups and assigns different groups to neighboring cells.



Frequency Reuse

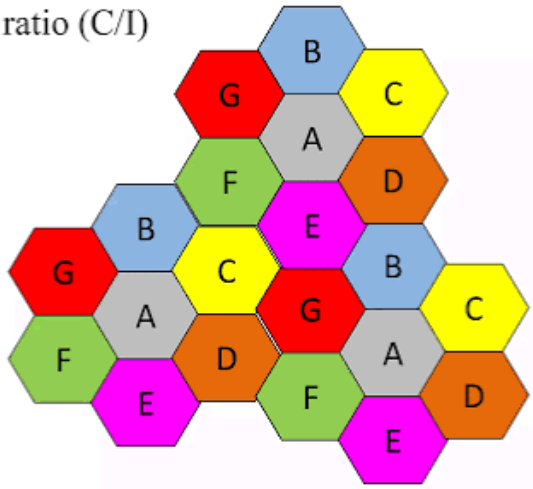
Frequency reuse is the technique of using the same frequency channels in multiple cells that are sufficiently separated geographically.

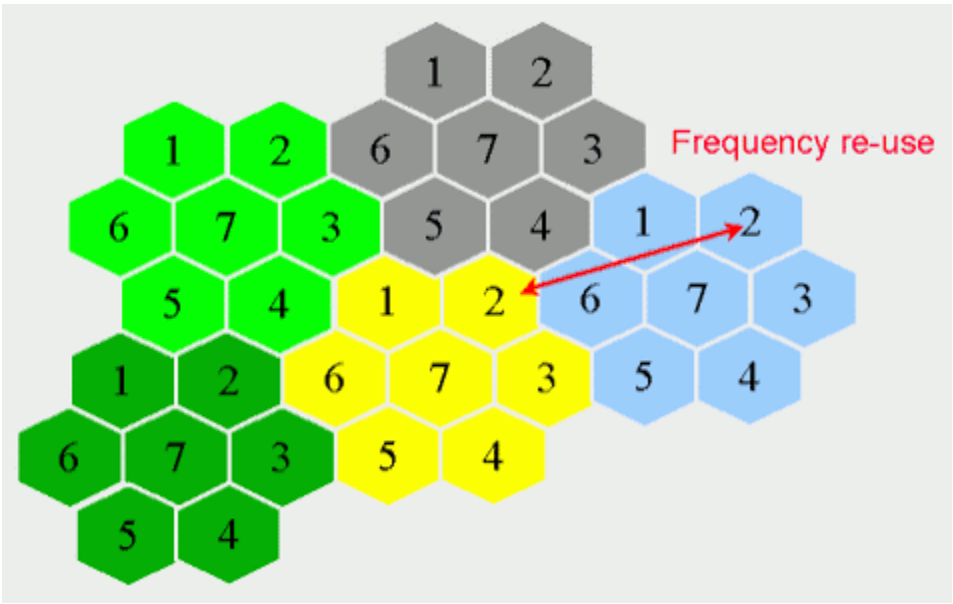
For example, suppose the available frequency groups are:

A, B, C, D, E, F, G

Frequency Reuse…

- Cells with same letter use the same set of frequency channels
- Using hexagonal cells, BS located at center of cell
- MS at edge of cell receives weak signal from BS, i.e., low Carrier to Interference ratio (C/I)





Cluster Size

A **cluster** is a group of cells in which each available frequency group is used only once.

The number of cells in a cluster is called the **cluster size**, represented by:

N

The commonly used cluster sizes are:

N=1,3,4,7,12,13,…

For this experiment, a **7-cell cluster** can be used because it clearly demonstrates frequency groups **A–G**.

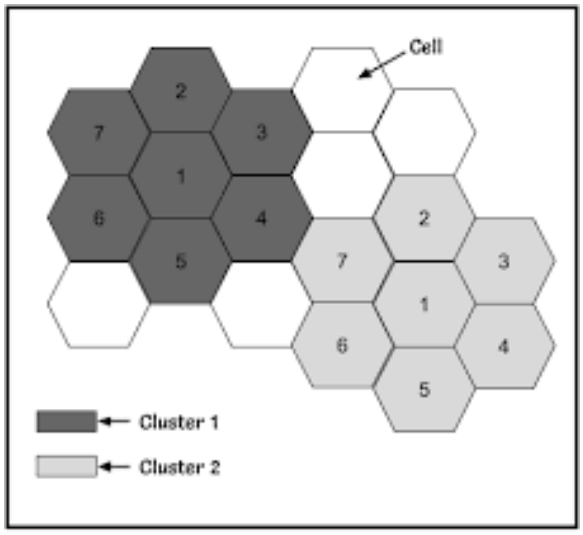


Figure: Cluster Size & Frequency Reuse

Frequency Reuse Distance

The distance between the centers of two nearest cells using the same frequency is called the **frequency reuse distance**, represented by D.

The relationship is:

RD=3N

where:

- D = frequency reuse distance
- R = radius of a cell
- N = cluster size

Therefore,

D=R3N

Example

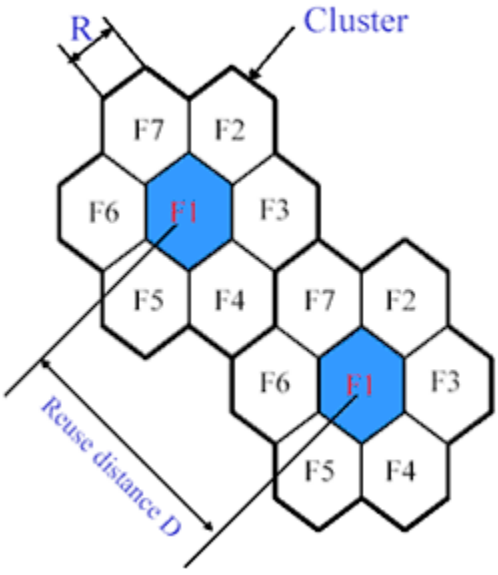
For a 7-cell cluster:

N=7

Therefore,

D=R21 D≈4.58R

Thus, two cells using the same frequency should be sufficiently separated to reduce interference.



For hexagonal cells, the reuse distance is given by

$$D = \sqrt{3N} \times R$$

where R is cell radius and N is the reuse pattern (the cluster size or the number of cells per cluster)

Reuse factor is

$$\frac{D}{R} = \sqrt{3 \times N}$$

Co-Channel Interference

When two cells use the same frequency channel, they are called **co-channel cells**.

If the co-channel cells are sufficiently far apart, interference can be controlled.

However, if the same frequency is assigned to adjacent cells, significant interference occurs.

Correct frequency reuse

```
   A
 B   C
D     E
 F   G
   A
```

The second **A** is sufficiently separated from the first **A**.

Incorrect frequency assignment

```
   A
 A   B
C     D
 E   F
   G
```

Execution Steps

Step 1: Open the Cellular Frequency Reuse Simulation

Open any cellular-network simulation or visualization software that allows you to create or view **hexagonal cellular networks**.

The simulator should ideally allow you to:

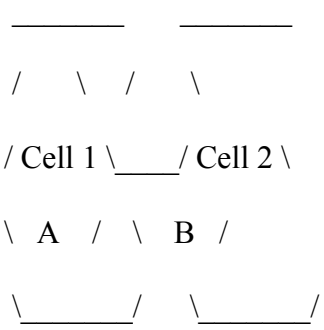
- Create multiple hexagonal cells
- Assign frequency groups to cells
- Change the frequency assigned to a cell
- Display the frequency-reuse pattern
- Observe interference between cells

What to observe

Initially, identify:

- Number of cells
- Base station location
- Cell boundaries
- Frequency groups
- Distance between cells

You should see a structure similar to:

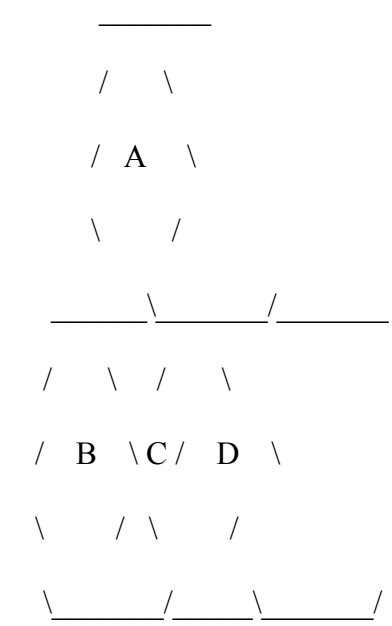


Step 2: Observe the Cellular Network

The cellular network is represented using **hexagonal cells**.

Each hexagonal cell represents a geographical coverage area served by one base station.

For example:



Each cell is given a different frequency group.

The **hexagonal shape** is commonly used in theoretical cellular planning because it provides a convenient approximation for circular radio coverage while allowing the cells to tile an area without gaps.

In the simulator, identify:

1. Cell boundaries
2. Base stations
3. Frequency assigned to each cell
4. Neighboring cells
5. Cells that are far enough apart to reuse frequencies

Step 3: Assign Frequency Groups A–G

Now assign seven different frequency groups to the cells in the first cluster.

For example:

Cell	Frequency Group
Cell 1	A
Cell 2	B
Cell 3	C
Cell 4	D
Cell 5	E
Cell 6	F
Cell 7	G

The first cluster therefore contains:

A, B, C, D, E, F, G

The important rule is:

Adjacent cells should not normally use the same frequency group.

For example:

[A]

[B] [C] [D]

[E] [F]

[G]

The actual arrangement will depend on the simulator.

Why different frequencies are used in adjacent cells?

If two neighboring cells use the same frequency, their signals can interfere strongly with each other.

Therefore, different frequency groups are initially assigned to neighboring cells.

Step 4: Reuse the Same Frequency Groups

After assigning A–G to the first cluster, create another cluster some distance away.

Now reuse the same frequency groups.

For example:

Cluster 1 Cluster 2

A B C D E F G A B C D E F G

| | | |

| | | |

+----------+-------+----------+

Sufficient reuse distance

For example:

- Cell 1 → A
- Cell 8 → A
- Cell 2 → B
- Cell 9 → B
- Cell 3 → C
- Cell 10 → C

and so on.

What is happening?

The frequency **A** is used again in another cell, but the two A cells are sufficiently far apart.

Because of the large separation, the interference between the two cells is kept within an acceptable level.

This is called **frequency reuse**.

Step 5: Observe and Identify the Reuse Pattern

Now observe the complete network.

Identify cells having the same frequency group.

For example:

Frequency Group	Cells Using It
A	Cell 1, Cell 8

B	Cell 2, Cell 9
C	Cell 3, Cell 10
D	Cell 4, Cell 11
E	Cell 5, Cell 12
F	Cell 6, Cell 13
G	Cell 7, Cell 14

You should visually identify that:

A → A

B → B

C → C

etc.

However, the repeated frequency cells should be separated by an appropriate **reuse distance**.

Frequency Reuse Distance

The distance between two cells using the same frequency is called the **frequency reuse distance**, generally represented by **D**.

For a hexagonal cellular system:

RD=3N

where:

- D = frequency reuse distance
- R = cell radius
- N = cluster size

For a cluster size of 7:

D=R21

Therefore:

D≈4.58R

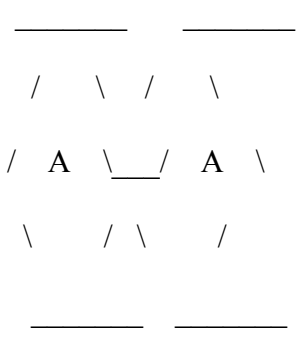
This means that cells using the same frequency should be sufficiently separated to reduce co-channel interference.

Step 6: Assign the Same Frequency to Adjacent Cells

Now perform the second part of the experiment.

Change the frequency assignment so that two neighboring cells use the **same frequency group**.

For example:



Here, two adjacent cells are using frequency **A**.

What should you observe?

You should observe increased **co-channel interference**.

The signal from Cell 1 using frequency A can interfere with the signal from the neighboring Cell 2, which is also using frequency A.

Conceptually:

Base Station 1

|

| A

↓

[Cell A]

↘

↘ Interference

↙

↙

[Cell A]

↑

| A

Base Station 2

Expected observations

You may observe:

- Signal distortion
- Reduced signal quality
- Increased interference
- Lower SINR
- Possible reduction in communication quality
- Increased error rate, depending on the simulator

The simulator may represent interference using different colors, signal-strength values, graphs, or numerical measurements.

Step 7: Record the Observations

Prepare an observation table such as the following.

Observation Table

Sr. No.	Frequency Assignment	Distance Between Same-Frequency Cells	Interference	Signal Quality
1	Different frequencies in adjacent cells	—	Very Low	Good
2	Same frequency in sufficiently distant cells	4.58R(Large)	Low/Acceptable	Good
3	Same frequency in moderately separated cells	2.50R(Medium)	Moderate	Reduced
4	Same frequency in adjacent cells	1.00R(Very small)	High	Poor
5	Seven frequency groups A–G reused in another cluster	Reuse distance(4.58R)	Acceptable	Good

If your simulator provides numerical values such as **SINR, received power, or interference power**, record those values as well.

For example:

Experiment Condition	Frequency	Interference Level	SINR
Adjacent cells, different frequencies	A/B	Low	High
Distant cells, same frequency	A/A	Low	High
Nearby cells, same frequency	A/A	High	Low

Code (1)-

```
import numpy as np
import matplotlib.pyplot as plt
import math

# ----------------------------
# PARAMETERS
# ----------------------------
R = 1
N = 7

# Frequency groups
frequencies = ['A', 'B', 'C', 'D', 'E', 'F', 'G']

# Frequency reuse distance
D = R * math.sqrt(3 * N)
```

```
print("FREQUENCY REUSE SIMULATION")
print("----------------------------")
print("Cell Radius (R) =", R)
print("Cluster Size (N) =", N)
print("Frequency Groups =", frequencies)
print("Frequency Reuse Distance (D) = %.2f R" % D)

# ----------------------------
# HEXAGON FUNCTION
# ----------------------------
def hexagon(x, y, radius):
    angles = np.arange(0, 2 * np.pi, np.pi / 3)
    return x + radius * np.cos(angles), y + radius * np.sin(angles)

# ----------------------------
# CREATE 19 CELLS
# ----------------------------
cells = []

for q in range(-2, 3):
    for r in range(-2, 3):
        if abs(q + r) <= 2:
            x = 1.5 * q
            y = math.sqrt(3) * (r + q / 2)
            cells.append((x, y))

# Sort cells by distance from center
cells.sort(key=lambda p: p[0]**2 + p[1]**2)

# Assign frequencies A-G repeatedly
assigned_frequency = {}

for i, cell in enumerate(cells):
    assigned_frequency[cell] = frequencies[i % 7]

# ----------------------------
# DRAW NETWORK
# ----------------------------
```

```
plt.figure(figsize=(10, 8))

for i, (x, y) in enumerate(cells):

    hx, hy = hexagon(x, y, R)

    freq = assigned_frequency[(x, y)]

    plt.fill(hx, hy, alpha=0.4)
    plt.plot(hx, hy, 'k-', linewidth=1)

    plt.text(
        x,
        y,
        freq,
        ha='center',
        va='center',
        fontsize=14,
        fontweight='bold'
    )

    # Base station
    plt.plot(x, y, 'ko', markersize=4)

plt.title("Frequency Reuse in Hexagonal Cellular Network")
plt.xlabel("Distance")
plt.ylabel("Distance")
plt.axis("equal")
plt.grid(True, alpha=0.3)

plt.show()

# ----------------------------
# FREQUENCY ASSIGNMENT TABLE
# ----------------------------
print("\nFREQUENCY ASSIGNMENT")
print("--------------------")

for i, cell in enumerate(cells):
```

```
    print(
        "Cell",
        i + 1,
        "-> Frequency",
        assigned_frequency[cell]
    )

# ----------------------------
# INTERFERENCE SIMULATION
# ----------------------------
print("\nINTERFERENCE OBSERVATION")
print("------------------------")

conditions = [
    ("Different frequencies - adjacent cells", 1.00, 0.05),
    ("Same frequency - distant cells", D, 0.10),
    ("Same frequency - moderately separated", 2.50, 0.35),
    ("Same frequency - adjacent cells", 1.00, 0.90)
]

print("\nCondition | Distance | Interference | SINR")

for condition, distance, interference in conditions:

    signal = 1.0
    noise = 0.05

    sinr = signal / (interference + noise)
    sinr_db = 10 * math.log10(sinr)

    print(
        f"{condition} | "
        f"{distance:.2f} R | "
        f"{interference:.2f} | "
        f"{sinr_db:.2f} dB"
    )

# ----------------------------
# FINAL OBSERVATION TABLE
```

```
# ----------------------------
print("\nFINAL OBSERVATION TABLE")
print("-----------------------")

print("\n1. Different frequencies in adjacent cells")
print("   Distance: 1.00 R")
print("   Interference: 0.05")
print("   Signal Quality: Good")

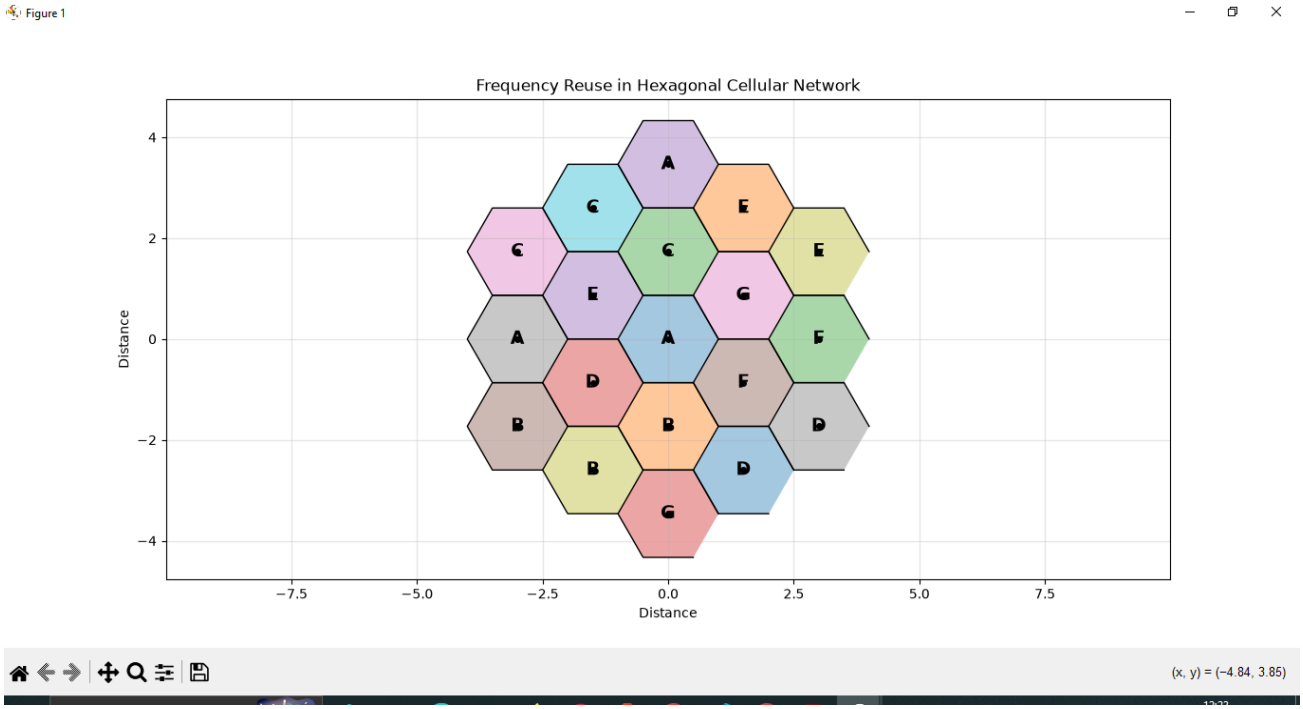
print("\n2. Same frequency in sufficiently distant cells")
print("   Distance: %.2f R" % D)
print("   Interference: 0.10")
print("   Signal Quality: Good")

print("\n3. Same frequency in moderately separated cells")
print("   Distance: 2.50 R")
print("   Interference: 0.35")
print("   Signal Quality: Reduced")

print("\n4. Same frequency in adjacent cells")
print("   Distance: 1.00 R")
print("   Interference: 0.90")
print("   Signal Quality: Poor")

print("\nCONCLUSION")
print("Frequency reuse improves spectrum efficiency by")
print("allowing the same frequencies to be reused in")
print("geographically separated cells while reducing")
print("co-channel interference.")
```

Output -



Code (2) -

```
import matplotlib.pyplot as plt
import numpy as np

R = 1

def draw_hex(x, y, label):
    angles = np.arange(0, 2 * np.pi, np.pi / 3)

    hx = x + R * np.cos(angles)
    hy = y + R * np.sin(angles)

    plt.fill(hx, hy, alpha=0.5)
    plt.plot(hx, hy, 'k-')

    plt.text(
        x,
        y,
        label,
```

```
        ha='center',
        va='center',
        fontsize=18,
        fontweight='bold'
    )

# Two adjacent cells using same frequency
draw_hex(0, 0, "A")
draw_hex(1.5, 0, "A")

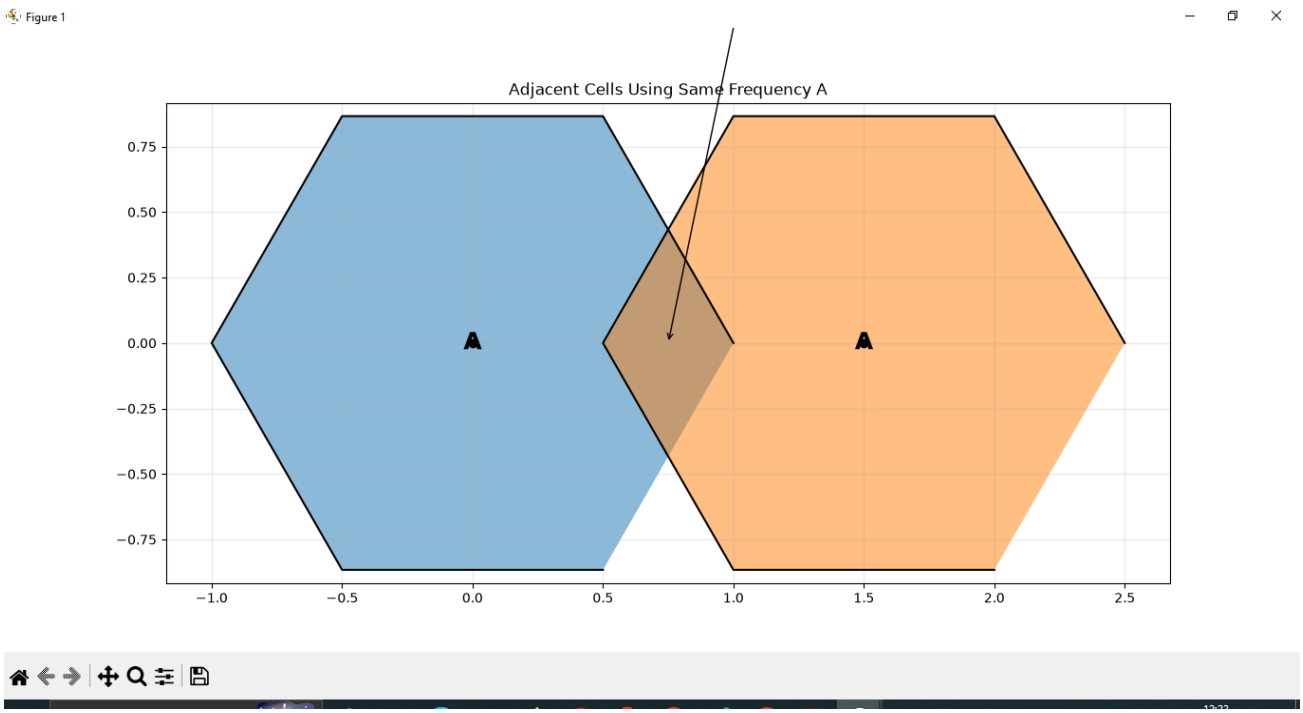
# Base stations
plt.plot(0, 0, 'ko')
plt.plot(1.5, 0, 'ko')

# Interference arrow
plt.annotate(
    "Co-channel Interference",
    xy=(0.75, 0),
    xytext=(0.75, 1.5),
    arrowprops=dict(arrowstyle="->")
)

plt.title("Adjacent Cells Using Same Frequency A")
plt.axis("equal")
plt.grid(True, alpha=0.2)

plt.show()
```

Output -



Conclusion:

The experiment demonstrates that frequency reuse is an essential technique in cellular communication systems for efficient utilization of limited radio-frequency spectrum.By dividing the service area into cells and reusing frequencies at an appropriate distance, a cellular network can serve a large number of users without requiring a separate frequency for every cell. However, improper frequency assignment, especially assigning the same frequency to adjacent cells, results in increased co-channel interference.